AHMAD NASER

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SUMMARY

Systems-focused Software Engineer with hands-on experience in Linux kernel internals, low-level C/C++ development, and OS-level performance tooling. Modified the Linux CFS scheduler on a custom Linux 6.6 kernel, built a ptrace-based syscall tracer, and implemented a custom heap allocator from scratch. Currently interning at Siraj Technologies on IoT system validation and E2E testing. Targeting kernel development, performance engineering, or systems software engineering roles

TECHNICAL SKILLS

Languages: C, C++, Python, Java, x86 Assembly, Bash
Kernel / OS: Linux, CFS (fair.c, core.c), ptrace, mmap, syscalls, paging, task_struct, vruntime, process scheduling, xv6, QEMU
Concurrency: Threads, mutexes, Peterson algorithm, sleep/wakeup, race condition debugging, IPC
Tooling: GDB, Valgrind, QEMU, Git, Make, Docker, Azure DevOps, AI-assisted tools (Claude, Copilot, Codex)
Backend: REST APIs, Spring Boot, TCP/IP, Sockets, BDD/Cucumber, Java/Maven, Angular

EDUCATION

B.Sc. in Computer Science — Ben-Gurion University of the Negev | GPA: 84.6 *2022 – 2025*
Relevant Courses: Operating Systems (C/C++), Data Structures, Algorithm Design, Compiler Principles, Programming Systems (C++, Java), Assembly

PROJECTS

SchedScope — Linux Kernel Scheduler Study *C · Linux 6.6 · QEMU · Python* SchedScope · Mini-Scheduler

- Modified CFS scheduler (**fair.c, core.c**) on a custom Linux 6.6 kernel in QEMU, introducing a configurable scheduling constant and observing its effect on task fairness and throughput under controlled multi-process workloads
- Measured scheduling behavior under load: modified policy produced **+64% context switches** and **+66% preemptions** with measurable fairness degradation (Jain's index drop), providing empirical grounding for scheduling policy trade-offs
- Built a deterministic user-space scheduler simulator (C) to reproduce Round Robin and CFS policies with identical trace formats for cross-environment validation
- Designed an end-to-end analysis pipeline (Python): kernel trace collection → metrics extraction → comparative evaluation

Syspeek — Syscall Tracing Tool *C · Linux · ptrace* GitHub

- Built a **ptrace**-based tracer (strace-equivalent) capturing syscall entry/exit, arguments, return values, and error codes across live processes; output format mirrors kernel-user-space interaction patterns
- Implemented multi-process tracing with signal forwarding, PTRACE_O_TRACEFORK support, and structured output that maps syscall numbers to names via kernel headers

HawkAlloc — Custom Heap Allocator *C · Linux · mmap* GitHub

- Implemented a heap allocator using the arena model with **mmap**, free-list (first-fit), forward coalescing, and block splitting; analyzed fragmentation vs. allocation speed trade-offs under varied workload patterns
- Validated correctness against glibc malloc behavior using custom stress tests; studied internal layout to understand alignment guarantees and metadata overhead

xv6 Kernel Extensions *C · RISC-V · QEMU* GitHub

- Extended the **xv6** kernel by implementing custom syscalls (forkn, waitall), virtual memory management
- Implemented shared memory IPC with a single-writer/multiple-readers model to ensure safe concurrent access
- Implemented spinlocks and sleep/wakeup, replacing busy-waiting with yield-based blocking to improve CPU utilization; validated via QEMU and GDB

EXPERIENCE

Software Engineering Intern — Siraj Technologies *03/2026 – Present*
IoT Smart Irrigation Platform (ADS) · Joint project with Netafim · Java/Maven · Angular · Docker · BDD

- Authored **BDD test scenarios using Cucumber** covering multi-component user flows across the full ADS stack (Angular frontend + Java backend + containerized services), improving system-level regression coverage
- Diagnosed cross-component defects by tracing interactions between Angular, Java/Maven services, and **Docker**-containerized infrastructure; reduced root-cause identification time using systematic log analysis and AI-assisted debugging (Claude, Copilot)
- Managed test pipelines and Git workflows using Azure DevOps across sprint cycles

VOLUNTEERING

2024: Peer mentor for undergraduate CS students in Data Structures and Combinatorics, Ben-Gurion University
2023: Introductory programming instructor for elementary school students

LANGUAGES

Arabic — Native · **Hebrew** — Fluent · **English** — Fluent